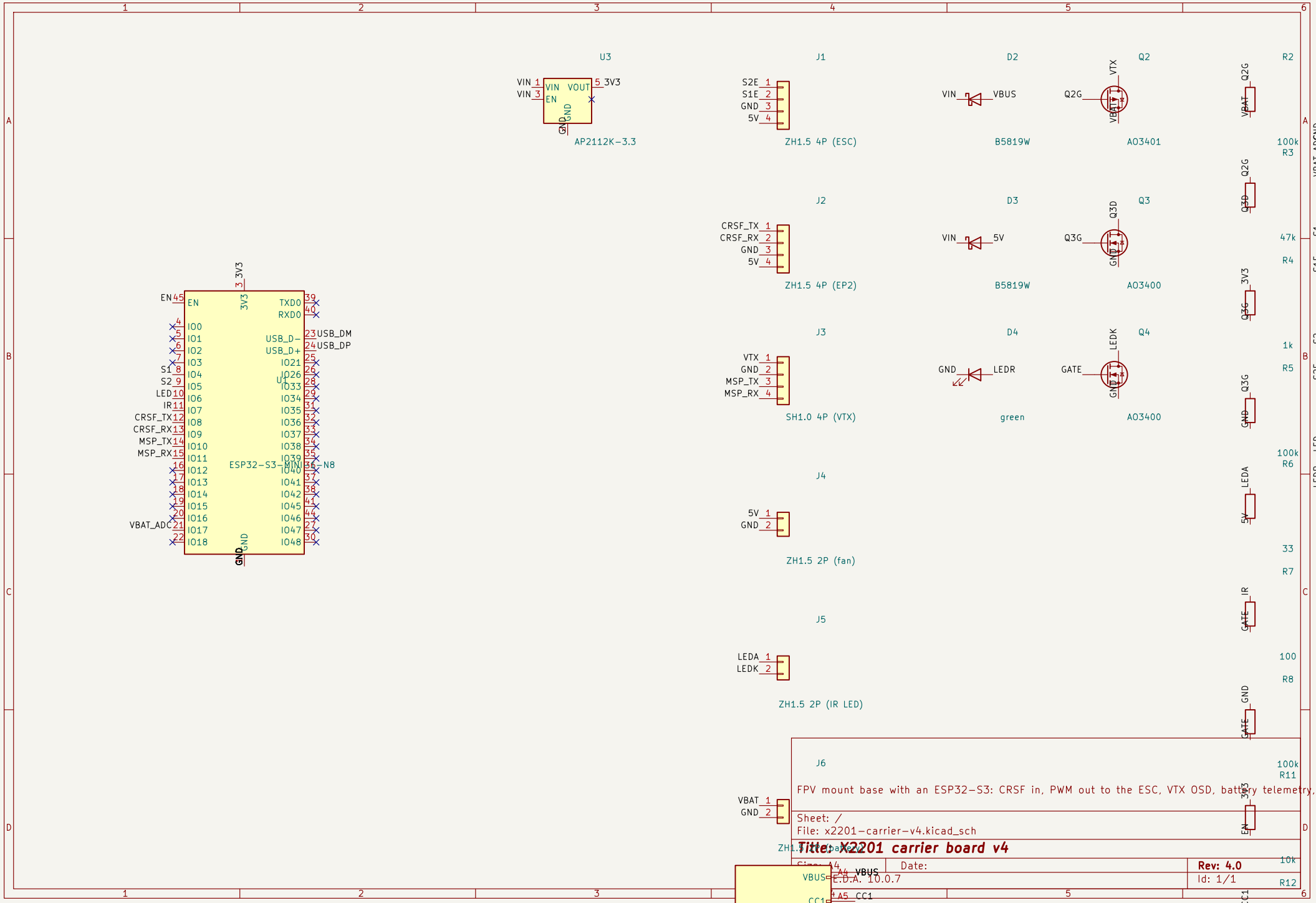

U3
VIN 1
VIN 3
VIN
EN
VOUT
5 3V3
GND
GND
AP2112K-3.3
J1
S2E 1
S1E 2
GND 3
5V 4
ZH1.5 4P (ESC)
D2
VIN
VBUS
B5819W
Q2
VTX
Q2G
VBAT
AO3401
R2
Q2G
VBAT
100k
R3
VBAT_ADC
J2
CRSF_TX 1
CRSF_RX 2
GND 3
5V 4
ZH1.5 4P (EP2)
D3
VIN
5V
B5819W
Q3
Q3D
Q3G
GND
AO3400
Q3D
Q3G
47k
R4
S1
S1E
J3
VTX 1
GND 2
MSP_TX 3
MSP_RX 4
SH1.0 4P (VTX)
D4
GND
LEDR
green
Q4
LEDK
GATE
GND
AO3400
Q3G
3V3
1k
R5
S2
S2E
J4
5V 1
GND 2
ZH1.5 2P (fan)
LED
LEDR
100k
R6
LEDA
5V
33
R7
J5
LEDA 1
LEDK 2
ZH1.5 2P (IR LED)
IR
GATE
100
R8
GND
GATE
J6
VBAT 1
GND 2
100k
R11
3V3
EN
10k
R12
CC1
GND
3V3
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
USB_D-
USB_D+
IO21
IO26
IO33
IO34
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
TXD0
RXD0
ESP32-S3-MINI-1-N8
U1
EN
S1
S2
LED
IR
CRSF_TX
CRSF_RX
MSP_TX
MSP_RX
VBAT_ADC
USB_DM
USB_DP
FPV mount base with an ESP32-S3: CRSF in, PWM out to the ESC, VTX OSD, battery telemetry, lap tra
Sheet: /
File: x2201-carrier-v4.kicad_sch
Title: X2201 carrier board v4
Size: A4
Date:
KiCad E.D.A. 10.0.7
Rev: 4.0
Id: 1/1
VBUS
CC1
CC2
D-
A4 VBUS
A5 CC1
B5 CC2
A7 USB_DM
B7 USB_DM